

Departement Elektriese, Elektroniese en  
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and  
Computer Engineering

## Semestertoets 1

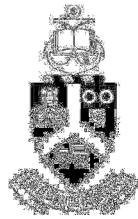
Kopiereg voorbehou

**Module EBN111**  
**Elektrisiteit en Elektronika**  
14 Maart 2013

## Semester Test 1

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**Module EBN111**  
**Electricity and Electronics**  
14 March 2013



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

### Eksamensinligting:

#### Exam information:

|                                                                                                        |                                |                                                  |                      |
|--------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------|----------------------|
| Maksimum punte:<br><i>Maximum marks:</i>                                                               | 50                             | Volpunte:<br><i>Full marks:</i>                  | 50                   |
| Duur van vraestel:<br><i>Duration of paper:</i>                                                        | 90 minute<br><i>90 minutes</i> | Oopboek / toeboek:<br><i>Open / closed book:</i> | Toe<br><i>Closed</i> |
| Totale aantal bladsye (hierdie blad ingesluit):<br><i>Total number of pages (including this page):</i> |                                | 17                                               |                      |

### BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.  
*The examination regulations of the University of Pretoria apply.*
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.  
*All questions must be answered on the ANSWER SHEET.*
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.  
*Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
- Antwoorde moet eenhede insluit!  
*Answers must include units!*
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**  
***Final answers must be written as real numbers, and not as fractions.***

|                                   |              |
|-----------------------------------|--------------|
| Dosent(e):<br><i>Lecturer(s):</i> | 1 J Joubert  |
|                                   | 2 JD le Roux |

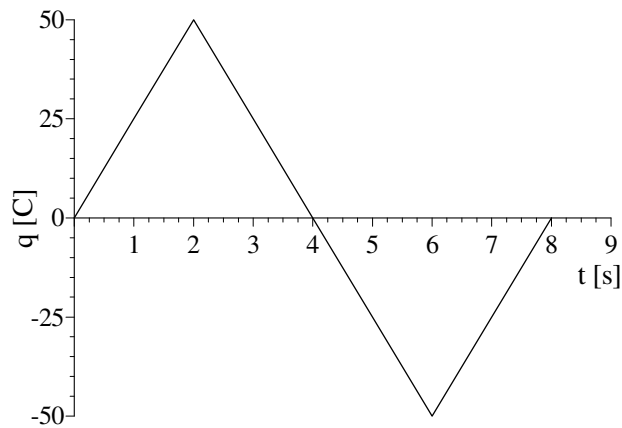
## VRAAG/QUESTION 1 [11]

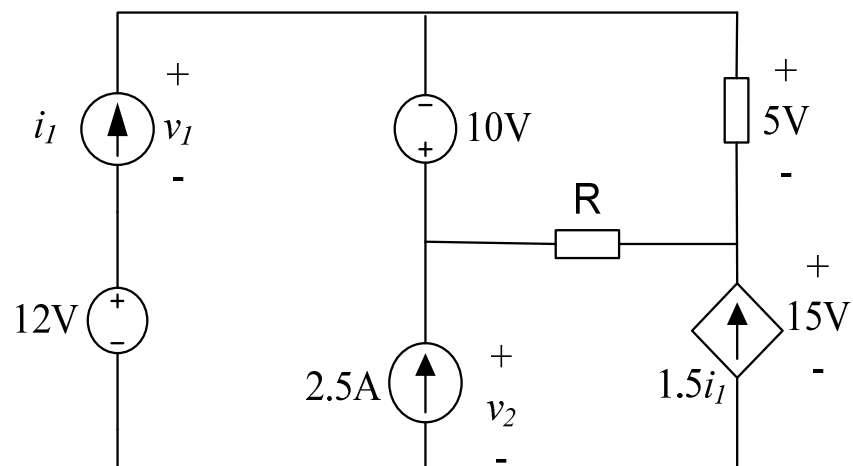
### **Vraag 1.1 [4]**

- a) 'n Gloeilamp absorbeer 100 Joule energie in 5 sekondes as dit by 'n 20 V toevoer ingeplug is. Hoeveel drywing absorbeer die gloeilamp.
- b) Jy gebruik 4 stoofplate om jou aandete voor te berei (plaat 1: 20 minute, plaat 2: 40 minute, plaat 3: 15 minute en plaat 4: 30 minute). Elke plaat gebruik 1.2 kW drywing. Bereken die totale energie (in kWh en kJ) wat jy gebruik het.
- c) Die lading wat in 'n draad vloei word hieronder getoon. Wat is die stroom in die draad by  $t = 4$  s.

### **Question 1.1 [4]**

- a) A light bulb absorbs 100 Joules of energy in 5 seconds if connected to a 20V supply. How much power is the bulb absorbing.
- b) You use 4 stove plates to prepare your dinner (plate 1: 20 minutes, plate 2: 40 minutes, plate 3: 15 minutes and plate 4: 30 minutes). Each stove plate uses 1.2 kW power. Calculate the total energy (in kWh and kJ) which you have used.
- c) The charge flowing in a wire is plotted below. What is the current in the wire at  $t = 4$  s.



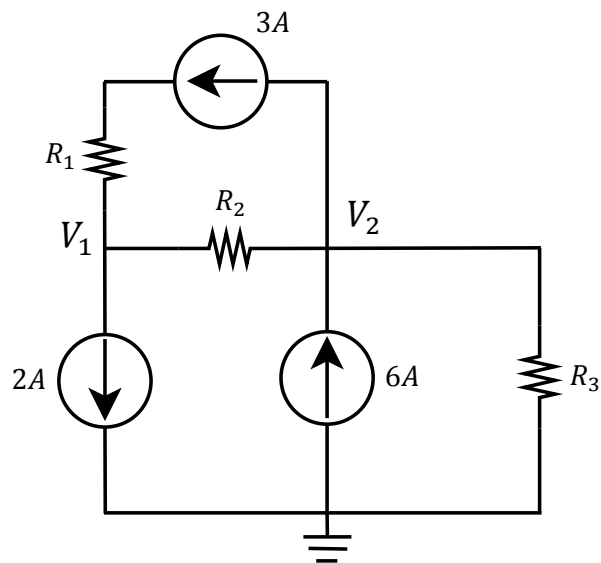
**Vraag 1.2 [3]**Bepaal die spannings  $v_1$ ,  $v_2$  en die stroom  $i_1$ .**Question 1.2 [3]**Determine the voltages  $v_1$ ,  $v_2$  and the current  $i_1$ .

**Vraag 1.3 [4]**

- a) Aanvaar dat die nodespannings in die stroombaan as volg is:  $V_1 = 500\text{V}$  en  $V_2 = 400\text{V}$ . Bereken die drywings ge-assosieer met die  $6\text{A}$  stroombron en die  $R_2$  weerstand.
- b) Indien die drywing ge-assosieer met die  $3\text{A}$  stroombron  $-450\text{W}$  is, bepaal die waarde van  $R_1$ .

**Question 1.3 [4]**

- a) Assume the node voltages in the circuit to be:  $V_1 = 500\text{V}$  and  $V_2 = 400\text{V}$ . Calculate the power associated with the  $6\text{A}$  current source and the  $R_2$  resistor.
- b) If the power associated with the  $3\text{A}$  current source is  $-450\text{W}$ , determine the value of  $R_1$ .



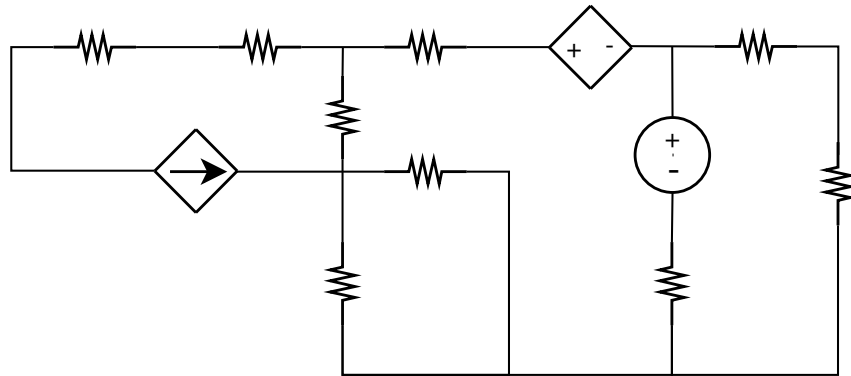
## VRAAG/QUESTION 2 [25]

### **Vraag 2.1 [4]**

- a) 'n Stroombaan het 5 takke en 3 onafhanklike lusse. Hoeveel nodusse is daar? [1]  
b) Bepaal hoeveel takke, nodusse en onafhanklike lusse daar in onderstaande stroombaan is. [3]

### **Question 2.1 [4]**

- a) A circuit has 5 branches and 3 independent loops. How many nodes are there? [1]  
b) Determine the number of branches, nodes and independent loops of the circuit below. [3]

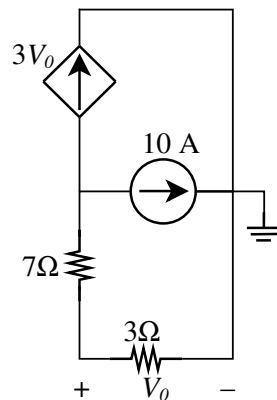


**Vraag 2.2 [3]**

Bepaal  $V_o$  [2] en die drywing [1] geassosieer met die onafhanklike stroombron.

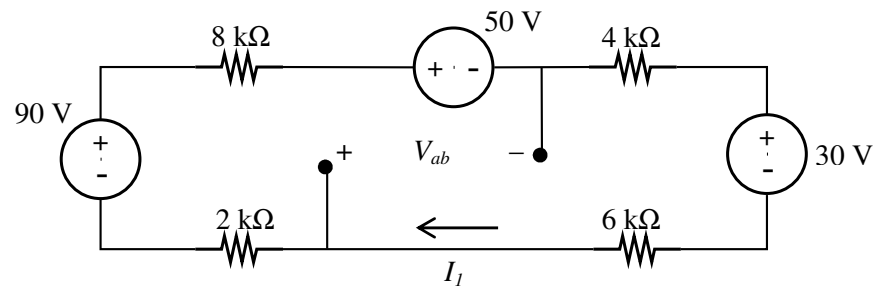
**Question 2.2 [3]**

Determine  $V_o$  [2] and the power [1] associated with the independent current source.



**Vraag 2.3 [2]**  
Bepaal  $I_I$  en  $V_{ab}$ .

**Question 2.3 [2]**  
Determine  $I_I$  and  $V_{ab}$ .

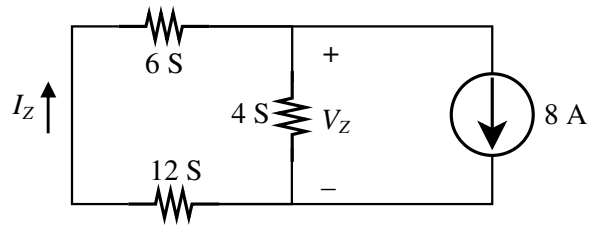


**Vraag 2.4 [3]**

Bepaal:

a)  $V_Z$  [2]b)  $I_Z$  [1]**Question 2.4 [3]**

Determine:

a)  $V_Z$  [2]b)  $I_Z$  [1]



**Vraag 2.5 [4]**

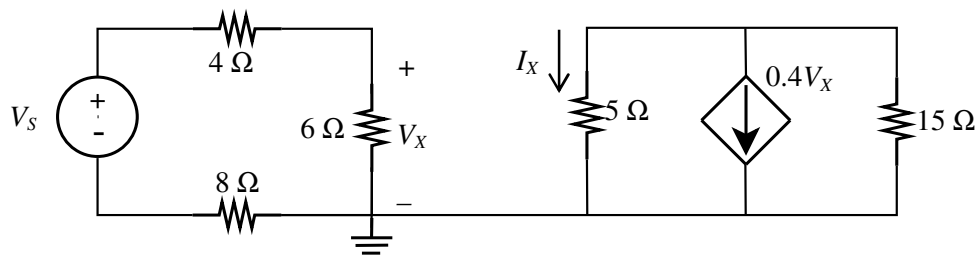
Bepaal:

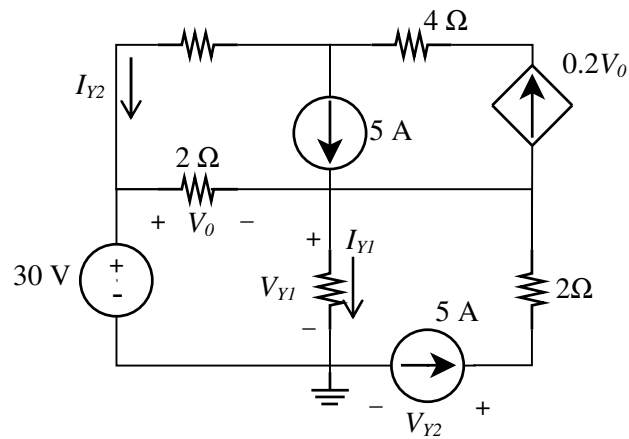
- a)  $V_X$  indien  $V_S = 9\text{ V}$ . [2]  
b)  $I_X$  indien  $V_X = 5\text{ V}$ . [2]

**Question 2.5 [4]**

Determine:

- a)  $V_X$  if  $V_S = 9\text{ V}$ . [2]  
b)  $I_X$  if  $V_X = 5\text{ V}$ . [2]



**Vraag 2.6 [4]**Indien  $V_0 = 10\text{ V}$ , bepaal:a)  $V_{Y1}$  en  $V_{Y2}$  [2]b)  $I_{Y1}$  en  $I_{Y2}$  [2]**Question 2.6 [4]**If  $V_0 = 10\text{ V}$ , determine:a)  $V_{Y1}$  and  $V_{Y2}$  [2]b)  $I_{Y1}$  and  $I_{Y2}$  [2]

**Vraag 2.7 [6]**

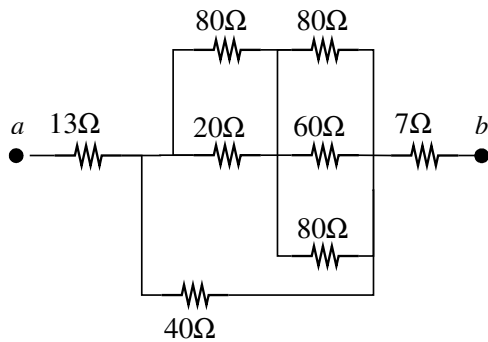
Bepaal die ekwivalente weerstand by terminale  $a$ - $b$  vir stroombane (1), (2) en (3).

(Stroombane (2) en (3) is op aparte bladsye)

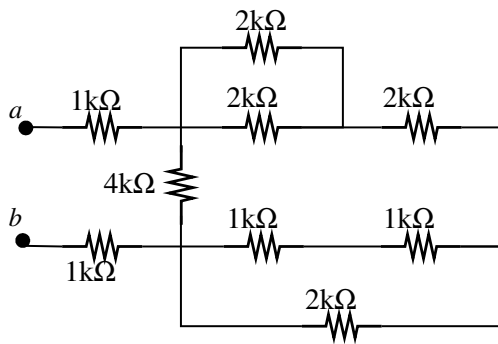
**Question 2.7 [6]**

Determine the equivalent resistance at terminals  $a$ - $b$  for circuits (1), (2) and (3).

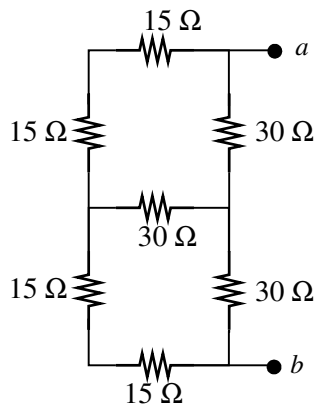
(Circuits (2) and (3) are on separate pages)

**Stroombaan/Circuit (1)**

### Stroombaan/Circuit (2)



**Stroombaan/Circuit (3)**



### VRAAG/QUESTION 3 [14]

**Vraag 3.1 [4]**

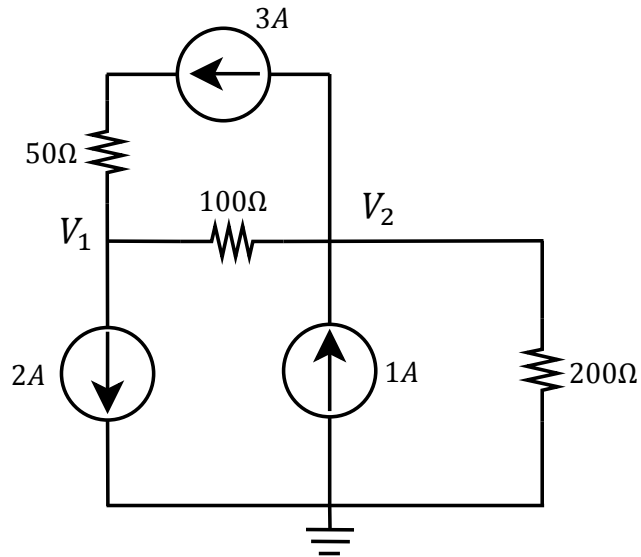
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} aV_1 + bV_2 &= 100 & (\text{node 1}) \\ cV_1 + dV_2 &= -400 & (\text{node 2}) \end{aligned}$$

**Question 3.1 [4]**

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} aV_1 + bV_2 &= 100 & (\text{node 1}) \\ cV_1 + dV_2 &= -400 & (\text{node 2}) \end{aligned}$$



**Vraag 3.2 [4]**

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} eV_1 + 5V_2 &= f \\ gV_1 + hV_2 &= 300 \end{aligned}$$

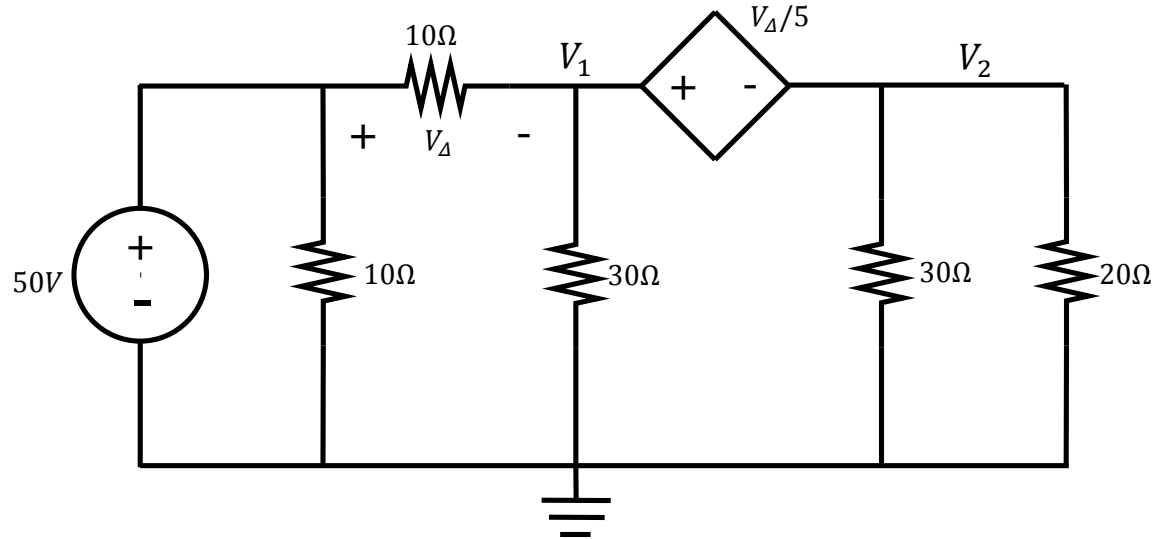
(wenk: die tweede vergelyking is opgestel mbv KCL)

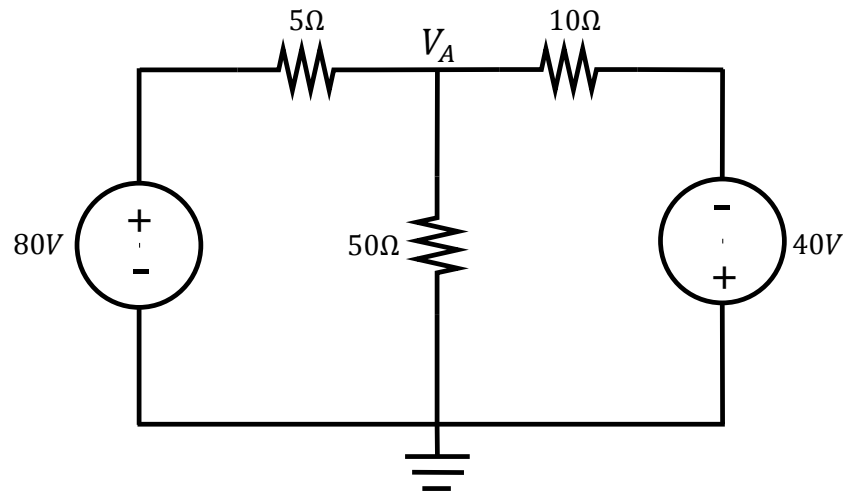
**Question 3.2 [4]**

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} eV_1 + 5V_2 &= f \\ gV_1 + hV_2 &= 300 \end{aligned}$$

(hint: the second equation was set using KCL)



**Vraag 3.3 [2]**Gebruik node-analise om node-spanning  $V_A$  te bereken.**Question 3.3 [2]**Use nodal analysis to calculate node voltage  $V_A$ .



**Vraag 3.4 [4]**a) Bepaal  $V_I$ .

$$\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

b) Bepaal  $i_3$ .

$$\begin{bmatrix} 1 & -3 & 0 \\ 3 & 2 & 2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

**Question 3.4 [4]**a) Determine  $V_I$ .

$$\begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

b) Determine  $i_3$ .

$$\begin{bmatrix} 1 & -3 & 0 \\ 3 & 2 & 2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

**EBN111 SEMESTER TOETS/TEST 1 (14 MAART/MARCH 2013)**  
**ANTWOORDBLAD / ANSWER SHEET**

|                             |  |                                    |  |  |  |     |     |  |  |
|-----------------------------|--|------------------------------------|--|--|--|-----|-----|--|--|
| Van / Surname:              |  | Studentenommer:<br>Student number: |  |  |  |     |     |  |  |
| Voorletters / Initials:     |  | Tel. nr. / Tel no.:                |  |  |  |     |     |  |  |
| Dissipline /<br>Discipline: |  | Taal / Language:                   |  |  |  | Afr | Eng |  |  |

|    |  |    |  |    |  |             |            |
|----|--|----|--|----|--|-------------|------------|
| Q1 |  | Q2 |  | Q3 |  | <b>Tot:</b> | <b>/50</b> |
|----|--|----|--|----|--|-------------|------------|

| VRAAG 1 – QUESTION 1 (11) |              |                 |                |           |
|---------------------------|--------------|-----------------|----------------|-----------|
| 1.1                       | $P =$        | $W = \quad kWh$ | $W = \quad kJ$ | $i(4s) =$ |
| 1.2                       | $v_I =$      | $v_2 =$         | $i_I =$        |           |
| 1.3                       | $P_{(6A)} =$ | $P_{(R2)} =$    | $R_I =$        |           |

| VRAAG 2 – QUESTION 2 (25) |                                              |                         |                         |
|---------------------------|----------------------------------------------|-------------------------|-------------------------|
| 2.1                       | a) Nodusse / Nodes =                         |                         |                         |
|                           | b) Onafhanklike lusse:<br>Independent loops: | Takke:<br>Branches:     | Nodusse:<br>Nodes:      |
| 2.2                       | $V_o =$                                      | $P =$                   |                         |
| 2.3                       | $V_{ab} =$                                   | $I_I =$                 |                         |
| 2.4                       | a) $V_Z =$                                   | b) $I_Z =$              |                         |
| 2.5                       | a) $V_X =$                                   | b) $I_X =$              |                         |
| 2.6                       | a) $V_{Y1} =$                                | $V_{Y2} =$              |                         |
|                           | b) $I_{Y1} =$                                | $I_{Y2} =$              |                         |
| 2.7                       | Circuit (1): $R_{eq} =$                      | Circuit (2): $R_{eq} =$ | Circuit (3): $R_{eq} =$ |

**VRAAG 3 – QUESTION 3 (14)**

|     |         |       |
|-----|---------|-------|
| 3.1 | $a =$   | $b =$ |
|     | $c =$   | $d =$ |
| 3.2 | $e =$   | $f =$ |
|     | $g =$   | $h =$ |
| 3.3 | $V_A =$ |       |
| 3.4 | $V_I =$ |       |
|     | $i_3 =$ |       |